

Program : Diploma in Integrated Circuit Design and Fabrication	
Course Code : 3378	Course Title: Simulation Lab
Semester : 3	Credits: 1.5
Course Category:	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To simulate and test mathematical and functional aspects of electronics, analog and digital electronic circuits using the available numerical computing software. (eg: MATLAB/SIMULINK, SCILAB/XCOS, PSPICE, GNU OCTAVE)

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic mathematics	1002 2002	Mathematics I,II	1&2
Characteristics of basic electronic circuits	2041 3043	Basic Electronics Electronic circuits	2 3
Implementation of combinational and sequential circuits	3044	Digital Electronics	3

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Develop programs to perform basic matrix operations, solve equations and plot basic signals	9	Applying
CO2	Develop simulations to demonstrate basic electronic circuits and study their characteristics	12	Applying
CO3	Develop simulation models to implement various combinational and sequential logic circuits.	9	Applying
CO4	Develop simulation models to implement low frequency and high frequency oscillator circuits.	9	Applying
	Lab Exam	6	Applying

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3					
CO2	3	3	3				
CO3	3	3	3				
CO4	3		3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Develop programs to perform basic matrix operations, solve equations and plot basic signals.		
M1.01	Perform basic matrix operations - addition, subtraction, multiplication, division and inverse (Verify mathematically)	3	Applying
M1.02	Solution of linear constant coefficient differential equation (Verify mathematically)	1	Applying
M1.03	Determine roots of a polynomial (Verify mathematically)	1	Applying
M1.04	Generate sine wave and cosine wave and plot in same window	1	Applying
M1.05	Plot different types of plots 3D, surface plot, polar plot	3	Applying
CO2	Develop simulations to demonstrate basic electronic circuits and study their characteristics		
M2.01	Plot input -output characteristics of diode	2	Applying
M2.02	Plot input -output characteristics of NPN BJT & MOSFET	3	Applying
M2.03	Design a single stage CE amplifier with potential barrier divider bias and 1) Plot its frequency response and 2) Measure the Bandwidth	3	Applying
M2.04	Build an emitter follower circuit and 1) Observe the frequency response 2) Measure the gain	2	Applying

	3) Plot its input / output waveform		
M2.05	Construct a single stage tuned amplifier circuit and 1) Plot input/output waveform 2) Plot its frequency response	2	Applying
	Lab Exam 1	3	Applying
CO3	Develop simulation models to implement various combinational and sequential logic circuits.		
M3.01	Simulate AND, OR, NAND, NOR, XOR, NOT Gates using blocks	2	Applying
M3.02	Develop a model of Full adder and full subtractor	2	Applying
M3.03	Develop a model of 3 bit multiplexer and demultiplexer.	2	Applying
M3.04	Develop a model of D, T, and JK Flip-flop using NAND gates.	3	Applying
CO4	Develop simulation models to implement low frequency and high frequency oscillator circuits		
M4.01	Design a RC phase shift oscillator and 1) Plot the output waveform 2) Measure the frequency of oscillation	1.5	Applying
M4.02	Design a Wein bridge oscillator and 1) Plot the output waveform 2) Measure the frequency of oscillation	1.5	Applying
M4.03	Design a Hartley oscillator and 1) Plot the output waveform 2) Measure the frequency of oscillation	1.5	Applying
M4.04	Design a Colpitts oscillator and 1) Plot the output waveform 2) Measure the frequency of oscillation	1.5	Applying
	Lab Exam 2	3	

Text / Reference:

T/R	Book Title/Author
T1	AgamkumarTyagi, MATLAB and Simulink for Engineers, Oxford Higher Education,2011
T2	Anil Kumar Verma, Scilab A Beginner's Approach, Cengage India, 2021

R1	Cleve Moler, Experiments with MATLAB, 2011
R2	Sandeep Nagar, Introduction to Scilab: For Engineers and Scientists Kindle Edition, Apress, 2017

Online resources:

Sl.No.	Website Link
1	https://www.mathworks.com/content/dam/mathworks/mathworks-dotcom/moler/exm/book.pdf
2	https://www.mccormick.northwestern.edu/documents/students/undergraduate/introduction-to-matlab.pdf
3	https://www.scilab.org
4	https://scilab.in/spoken-tutorial